1.1: Problem

People who enjoy being outdoors usually don't know if the air is safe for walking or running until they check it up online, which usually involves opening an app or a website and searching for their position each time, that is why most people end up not doing it. Since smoke from wildfires and regular pollution can get anyone sick, it shouldn't take a lot of effort to find out about the air quality.

1.2: Solution

My solution is a Java command line tool which finds the current air quality for any location from the free Open Meteo Air Quality API and explains whether it is a good day to go outside. It can also give specific time periods to go outside and do different activities. If the air quality goes below the level you select, it displays a clear warning together with a quick advice on how to stay healthy.

2.2: Tech Stack

- Java 25: the programming language for the whole project
- IO.readln: reads user inputs
- Open Meteo Air Quality API: provides the current air quality data
- IP-API: finds the user’s approximate location
- java.time: handles the time each air quality reading was taken
- java.net.http.HttpClient: lets the program talk to websites and web APIs
- com.google.gson.*: turns the JSON text returned by the APIS into objects so I can get the AQI and coordinates easily out of them

2.3 Chunked Build Plan

- Set up Git repo and environment
- Get the program to get the user's location from IP-API
- Parse the coordinates from the IP-API response with Gson
- Get air quality data from the Open Meteo API using those coordinates
- Get the AQI number from the Open Meteo data
- Translate the AQI into english words
- Let the user set a warning level and show warnings when the air is worse
- Add error handling for no internet or failed API calls

3.1 - Repo Setup

https://github.com/yatharthjag2008/Air_Quality_Project